$$\text{adj}\,A = \begin{bmatrix} +\begin{vmatrix} 3 & 1 \\ 2 & 3 \end{vmatrix} & -\begin{vmatrix} -5 & 1 \\ -3 & 3 \end{vmatrix} & +\begin{vmatrix} -5 & 3 \\ -3 & 2 \end{vmatrix} \\ -\begin{vmatrix} -1 & 3 \\ 2 & 3 \end{vmatrix} & +\begin{vmatrix} 2 & 3 \\ -3 & 3 \end{vmatrix} & -\begin{vmatrix} 2 & -1 \\ -3 & 2 \end{vmatrix} \\ +\begin{vmatrix} -1 & 3 \\ 3 & 1 \end{vmatrix} & -\begin{vmatrix} 2 & 3 \\ -5 & 1 \end{vmatrix} & +\begin{vmatrix} 2 & -1 \\ -5 & 3 \end{vmatrix} \end{bmatrix}^T = \begin{bmatrix} 7 & 12 & -1 \\ 9 & 15 & -1 \\ -10 & -17 & 1 \end{bmatrix}^T = \begin{bmatrix} 7 & 9 & -10 \\ 12 & 15 & -17 \\ -1 & -1 & 1 \end{bmatrix}.$$

$$\text{Hence, } A^{-1} = \frac{1}{|A|}(\text{adj}\,A) = \frac{1}{(-1)}\begin{bmatrix} 7 & 9 & -10 \\ 12 & 15 & -17 \\ -1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} -7 & -9 & 10 \\ -12 & -15 & 17 \\ 1 & 1 & -1 \end{bmatrix}.$$

1.2.3 Properties of inverses of matrices

We state and prove some theorems on non-singular matrices.

Theorem 1.4

If A is non-singular, then

(i) $|A^{-1}| = \frac{1}{|A|}$ (ii) $(A^T)^{-1} = (A^{-1})^T$ (iii) $(\lambda A)^{-1} = \frac{1}{\lambda}A^{-1}$, where λ is a non-zero scalar.

Proof

Let A be non-singular. Then $|A| \neq 0$ and A^{-1} exists. By definition,

$$AA^{-1} = A^{-1}A = I_n. \qquad \ldots(1)$$

(i) By (1), we get $|AA^{-1}| = |A^{-1}A| = |I_n|$.

Using the product rule for determinants, we get $|A||A^{-1}| = |I_n| = 1$.

Hence, $|A^{-1}| = \frac{1}{|A|}$.

(ii) From (1), we get $(AA^{-1})^T = (A^{-1}A)^T = (I_n)^T$.

Using the reversal law of transpose, we get $(A^{-1})^T A^T = A^T (A^{-1})^T = I_n$. Hence $(A^T)^{-1} = (A^{-1})^T$.

(iii) Since λ is a non-zero scalar, from (1), we get $(\lambda A)\left(\frac{1}{\lambda}A^{-1}\right) = \left(\frac{1}{\lambda}A^{-1}\right)(\lambda A) = I_n$.

So, $(\lambda A)^{-1} = \frac{1}{\lambda}A^{-1}$.

Theorem 1.5 (Left Cancellation Law)

Let $A, B,$ and C be square matrices of order n. If A is non-singular and $AB = AC$, then $B = C$.

Proof

Since A is non-singular, A^{-1} exists and $AA^{-1} = A^{-1}A = I_n$. Taking $AB = AC$ and pre-multiplying both sides by A^{-1}, we get $A^{-1}(AB) = A^{-1}(AC)$. By using the associative property of matrix multiplication and property of inverse matrix, we get $B = C$.

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